

Ishan Apte

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Summary

AI Engineer with expertise in developing machine learning solutions and implementing production-ready AI systems. Proficient in deep learning frameworks, natural language processing, and MLOps practices. Demonstrated success in building scalable AI applications through multiple projects and research experience. Skilled in translating complex AI concepts into practical solutions while collaborating with cross-functional teams.

Education

Indiana University Bloomington

MS in Computer Science

Aug 2023 - May 2025

Bloomington, IN

Pune Institute of Computer Technology

Bachelor's in Computer Engineering, Minors in CyberSecurity

Aug 2019 - May 2023

Pune, India

Technical Skills

Programming Languages: Python, JavaScript, C++, SQL, TypeScript, Bash/Shell

AI/ML Frameworks & Libraries: TensorFlow, PyTorch, Keras, LangChain, Hugging Face Transformers

Cloud & MLOps: AWS (SageMaker, EC2, S3), Docker, Kubernetes, MLflow, Git, CI/CD, TensorBoard

Databases & Data Processing: PostgreSQL, MySQL, MongoDB, Redis, Apache Spark, ETL Pipelines

AI/ML Concepts: Deep Learning, NLP, Large Language Models, RAG, Transfer Learning, Model Deployment, Neural Networks

Experience

Software Developer

Indiana University Bloomington

Jun 2025 - Present

USA

- Built an iOS event discovery app using **SwiftUI**, implementing 5 key features including interactive mapping (**MapKit**) and location-based recommendations (**CoreLocation**), serving over **40,000** university students.
- Conducted user testing with 20+ peers using **agile methodology**, incorporating feedback that enhanced user experience and **reduced navigation time by 40%**.

Machine Learning Engineer

Pune Institute of Computer Technology

Jan 2023 - Aug 2023

Pune, India

- Trained a Convolutional Neural Network for weather condition classification, **achieving 88% accuracy** by processing over 5,000 weather images, contributing key findings that improved project outcomes by 30%.
- Optimized model performance through **grid search hyperparameter tuning** (learning rate, batch size, dropout rate) and implemented **k-fold cross-validation**, ensuring **15% improved reliability** across diverse weather scenarios.
- Implemented **TensorFlow**-based solutions following best practices in machine learning development, enhancing the research team's capabilities.

Projects

BrainstormAI 🧠 | *Node.js, LangChain.js, LangGraph, RESTful APIs*

- Engineered the system to **reduce AI pandering** and simulate the candid, creative energy of brainstorming with friends, fostering an authentic and collaborative environment that moves beyond simple conversational assistance.
- Architected a **stateful agent** using **LangGraph** to create **cyclical and conditional workflows**, enabling the AI to dynamically select the appropriate tool—such as response generation, note-taking, or question suggestion—based on conversational context.
- Utilized the **LangChain.js framework** to manage **conversational memory** and structure interactions with **large language models**, improving the coherence and context-awareness of the AI assistant.

AgendaLink | *Python, Node.js, NLP (GPT-3.5), RESTful APIs, OAuth2*

- Implemented advanced **Natural Language Processing (NLP)** using OpenAI's **GPT-3.5** for **intent detection**, processing user queries with **85% accuracy** across 7 distinct intent categories.
- Built cross-service integration using Node.js to extract event details from emails with OpenAI's API, demonstrating practical application of **Retrieval Augmented Generation (RAG)**.
- Designed system with an **API-first, microservice architecture** decoupling core agent logic from task execution components, demonstrating strong software architecture principles.

EventCraft 🧰 | *Python, Case-Based Reasoning, Tkinter, Pandas*

- Engineered a **Case-Based Reasoning (CBR)** system for IMU Catering, **automating logistical planning** for 15+ monthly events, reducing planning time by 35% through **adaptation of historical data**.
- Constructed three comprehensive **knowledge bases** (CSVs) using **pandas DataFrames**, enabling efficient **case retrieval** and **solution adaptation** within the CBR framework.
- Created an intuitive **Tkinter UI** for event parameter input, interfacing with the CBR engine to generate tailored plans and improve **user experience**.